

# High-Performance Star-M SVD for Big Data Compression


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Paper · Software

## Tensor Decomposition Landscape

Many high-dimensional scientific datasets are modeled as multidimensional arrays (tensors). Low-rank tensor decompositions make it feasible to store and operate on large data.

| Method | Optimality | HPC Ready |
|---|---|---|
| Matrix SVD | Exact (Eckart–Young) | ✓ |
| CP | None | ✓ |
| Tucker | $\sqrt{d}\times$ optimal | ✓ |
| Tensor Train | $\sqrt{d}\times$ optimal | ✓ |
| $\star_{\mathbf{M}}$ t-SVD | Exact (Eckart–Young) | ✗ |

**Goal:** compress GB–TB scientific data with provable optimality. $\star_{\mathbf{M}}$ t-SVD is exactly optimal, yet has no HPC solution.

## The Software Gap for $\star_{\mathbf{M}}$ t-SVD

**Existing**
MATLAB & pure Python

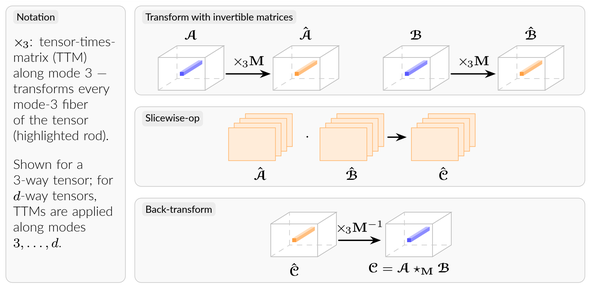
- ✗ Single-threaded
- ✗ Limited parallelism
- ✗ GB–TB data infeasible
- ✗ Small experiments only

**`pystarm` [3]**
C++ + Python

- ✓ Shared-memory parallel
- ✓ Utilizes all available cores
- ✓ GB–TB datasets
- ✓ Python interface

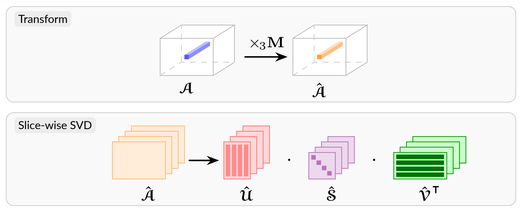
1000× faster than currently available solutions on a ≈3 GB Computational Fluid Dynamics (CFD) tensor

## $\mathcal{A} \star_{\mathbf{M}} \mathcal{B}$: The $\star_{\mathbf{M}}$ Framework [1]




The choice of $\mathbf{M}$ shapes the transform domain — a good $\mathbf{M}$, such as the Discrete Cosine Transform (DCT) matrix, concentrates information and improves compression.

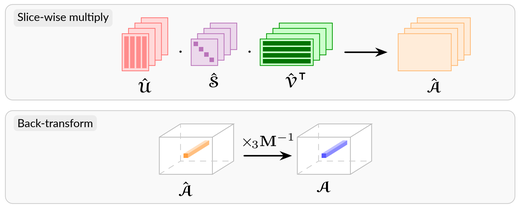
## The $\star_{\mathbf{M}}$ t-SVD Algorithm




$\star_{\mathbf{M}}$ t-SVD = TTM + $n$ **independent** matrix SVDs

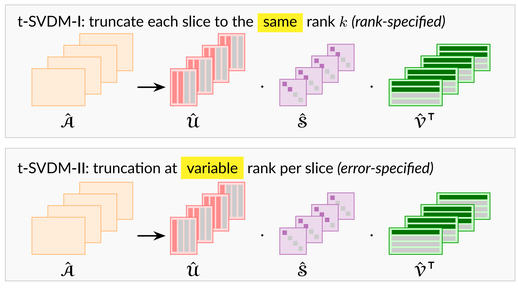
$$\hat{A}^{(i)} = \hat{U}^{(i)}\,\hat{S}^{(i)}\,\hat{V}^{(i)\intercal}, \quad i = 1, \ldots, n$$

## $\star_{\mathbf{M}}$ t-SVD Reconstruction



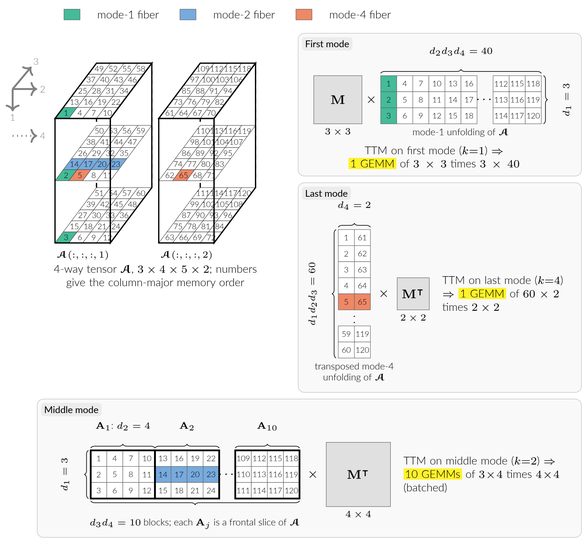

For $d$-way tensors: all modes beyond the 2nd are transformed — either mode by mode via individual TTMs, or equivalently via a single TTM with a Kronecker product transform. Slices are then treated identically.

## Two Truncation Strategies




Variable rank truncation opens up scope for **better compression**

## t-SVDM Workhorse-1: TTM

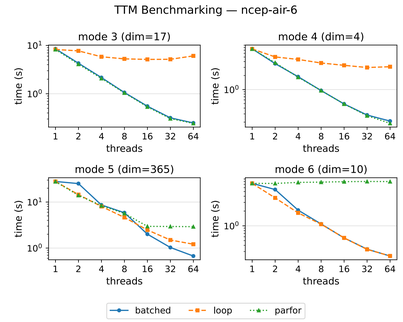



**Possible designs** (all three call Intel MKL)

- *loop* — loop over multithreaded GEMM
- *parfor* — parallel-for over sequential GEMM
- *batched* — batched strided GEMM, available in MKL

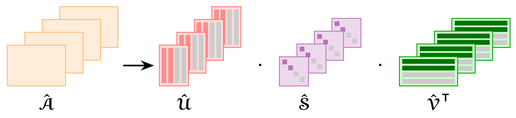
## TTM Benchmark Results

`ncep-air-6`: $73 \times 144 \times 17 \times 4 \times 365 \times 10$




Gaps follow from the batch count $P_k$. In **mode 3**, 14,600 GEMMs of $10{,}512 \times 17$ are too small for multithreaded GEMM (*loop*); in **mode 5**, $P_k = 10$, so *parfor* uses ≤10 threads; in **mode 6**, $P_k = 1$, so *parfor* is one sequential GEMM. *batched* is best.

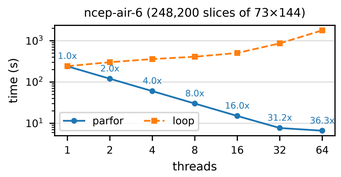
## t-SVDM Workhorse-2: Slice-wise SVD



**Possible designs**

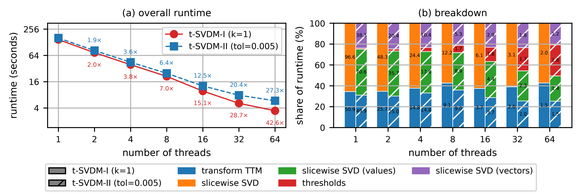
- **parallel slices, sequential SVD** — embarrassingly parallel, no dependencies between slices
- **sequential slices, parallel SVD**
- **a mix** — batched SVD, available in recent MKL but not as reliable

## Slice-wise SVD Benchmark Results




**Parallel slices, sequential SVD** scales near-linearly up to 64 threads; **sequential slices, parallel SVD** does not scale well, especially for small slices.

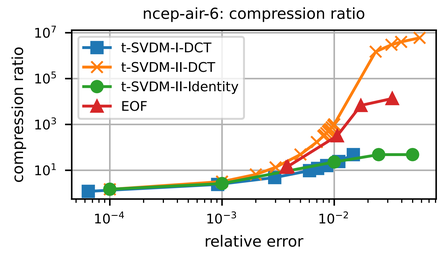
## Strong Scaling: `ncep-air-6`




t-SVDM-II at tol 0.005 (CR ≈ 48.9) and t-SVDM-I at rank $k = 1$ (CR ≈ 48.2). (a) total runtime vs. thread count, with speedup annotated; (b) per-stage breakdown as a percentage of the total.

## Compression: Climate Reanalysis Data

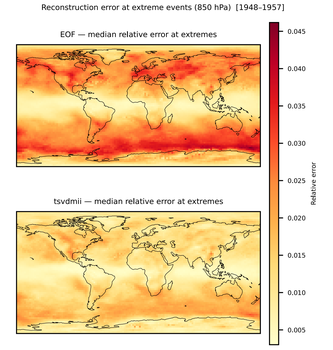
`ncep-air-6`: $73 \times 144 \times 17 \times 4 \times 365 \times 10$ (≈ 19.5 GB), $\mathbf{M}$: DCT




- **Tensor vs. matrix** — tensor methods outperform EOF (Empirical Orthogonal Functions) [2], the standard in climate science
- **t-SVDM-II vs. t-SVDM-I** — variable-rank truncation gives higher compression at every error level
- **DCT vs. identity** — DCT concentrates energy into fewer singular values per slice

## Extreme Event Preservation: t-SVDM-II vs. EOF

EOF, the standard practice in climate science, tends to wash out **extreme events**.




Median pointwise relative error at extreme temperature events (top/bottom 0.5% at 850 hPa) at a **matched compression ratio of** ≈320×. t-SVDM-II with DCT transform achieves **lower error across most of the globe**.

## Conclusions and Future Work

- 64-thread speedup: t-SVDM-I 42×, t-SVDM-II 27× — a good avenue to exploit parallel performance for tensor decomposition
- Tensor methods beat EOF and better preserve climate extremes
- Next: distributed memory, and applying parallel t-SVD in scientific simulations
- Code: `https://github.com/Streaming-Tensors/pystarm/`

### References


[1] M. E. Kilmer, L. Horesh, H. Avron, and E. Newman. Tensor-tensor algebra for optimal representation and compression of multiway data. *PNAS*, 118(28):e2015851118, 2021.

[2] A. Hannachi, I. T. Jolliffe, and D. B. Stephenson. Empirical orthogonal functions and related techniques in atmospheric science: A review. *Int. J. Climatol.*, 27(9):1119–1152, 2007.

[3] M. T. Hussain, G. Ballard, A. Devarakonda, S. Eswar, N. Pesricha, and V. Rao. High-performance Star-M SVD for big data compression. In *Proc. ICPP '26*, 2026. doi:10.1145/3832810.3832889.


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